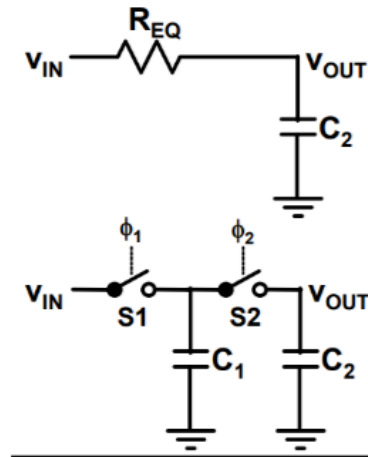


Switched Capacitor Background



- Switched cap resistor exhibits equivalent resistance of $R_{EQ} = \frac{T_S}{C_1}$, where T_S is the sampling period and C_1 is the sampling capacitor

- Example:

$$T_S = 1\mu s, C_1 = 1pF$$

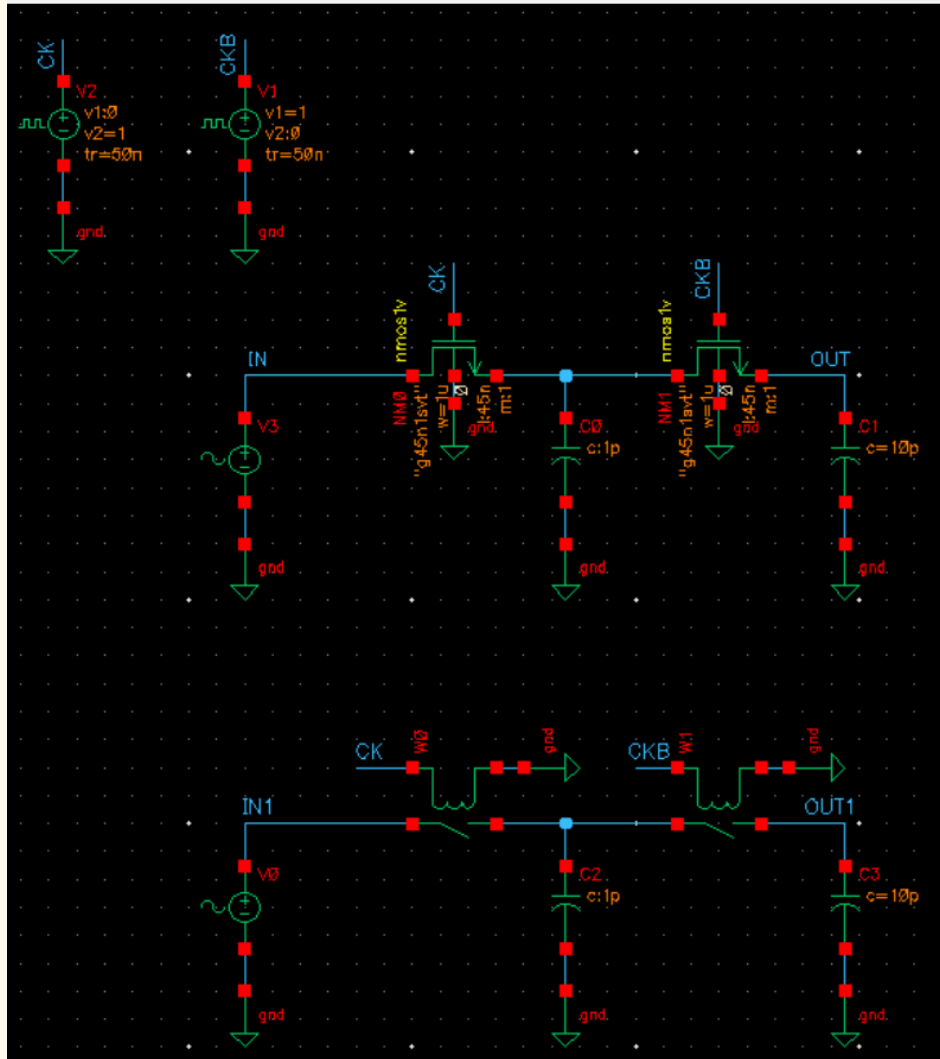
$$R_{EQ} = \frac{T_S}{C_1} = 1M\Omega$$

- For passive switched cap low pass filter with $R_{EQ} = 1M\Omega$ and $C_2 = 10pF$, the 3dB cut-off frequency is 15.6kHz

- from the driving point concept, the switched capacitor looks like a resistor
- using the sampling time and capacitor
 - on average it always drawing some current for some voltage, and the ratio relates to the T_S and C
 - and it looks like a resistor
- When C is big. the applying voltage will need to pass more charge, so larger current, so smaller R
- When T_S is longer, it takes longer to charge, so R is larger
- No need to use real resistor to make filter
 - R and C are both $\pm 20\%$, so cutoff is $\pm 40\%$, this is big
 - Using the switched capacitor, the BW of filter is related to $RC = T_S/C_1$ $C_2 = T_S C_2/C_1$
 - clock is typically very accurate, so no worries with T_S

- $C_2/C_1 \Rightarrow$ is a ratio, good design
- make the cap bigger to get even better matching

Cadence Setup

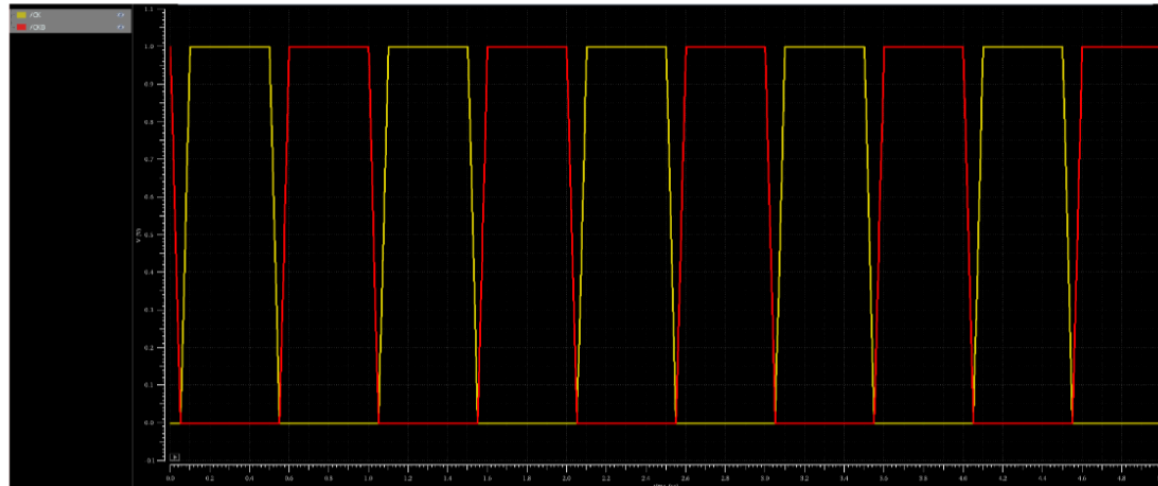


- Create the passive switched cap lowpass filter as shown. All transistors have length of 45nm and width of 1 μ m
- C0 and C2 are the sampling capacitor of 1pF while C1 and C3 are the capacitors for RC lowpass filter of 10pF
- Ideal switch (W0 and W1) with “open voltage” of 0.4V and “close voltage” of 0.6V exhibit open resistance of 1T Ω and close resistance of 1 Ω . This allows us to compare the performance between ideal switched cap circuit and the one using actual transistor as switch

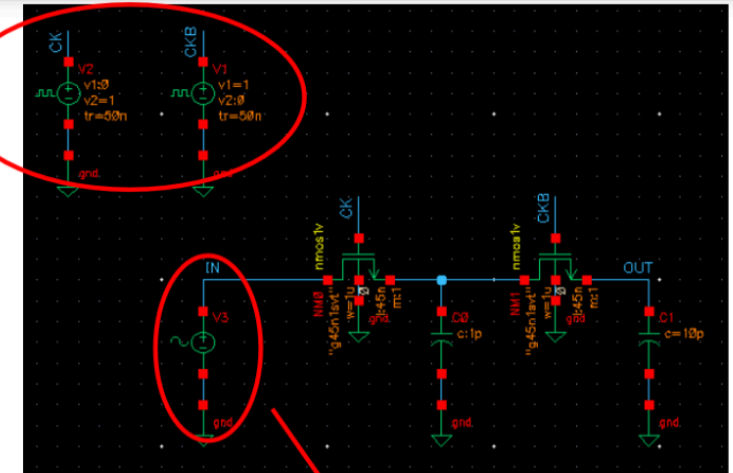
- the ideal switch is in analog lib

Component Setup

PAC phase		PAC phase	
Voltage 1	0 V	Voltage 1	1 V
Voltage 2	1 V	Voltage 2	0 V
Period	1 u s	Period	1 u s
Delay time	50 n s	Delay time	
Rise time	50 n s	Rise time	50 n s
Fall time	50 n s	Fall time	50 n s
Pulse width	400 n s	Pulse width	500 n s
Temperature coefficient 1		Temperature coefficient 1	



- Use vsin source for V3 and V0 source and define PAC magnitude
- Use vpulse source for CK and CKB and define the parameters as shown to create non-overlapping clock



Edit Object Properties

User Property: lvsignore TRUE

CDF Parameter	Value
First frequency name	
Second frequency name	
Noise file name	
Number of noise/freq pairs	0
DC voltage	
AC magnitude	
AC phase	
XF magnitude	
PAC magnitude	1 V
PAC phase	
Delay time	

OK Cancel Apply Defaults

- CKB is antiphase, B for bar
- Need to make sure the clock don't overlap, so the two switches won't short through
 - plot in time to check
- PAC for periodic simulation

Setup PSS Simulation

- Select [Analyses->Choose...](#)
- Choose pss analysis and check “Auto Calculate”. The “Beat Frequency” should indicate 1MHz, which is the same as our sampling clock
- Define “Output harmonics” to be 6. This will tell the simulator to simulate the modulated input effect up to 6 harmonics
- Check “moderate” under “Accuracy Defaults”
- Check “Yes” for “Run transient”
- Check “Detect Steady State”, and define Stop Time (tstab) to be 5u

Choosing Analyses -- ADE Explorer

Analysis

☐ tran ☐ dc ☐ ac ☐ noise

☐ xf ☐ sens ☐ dcmatch ☐ acmatch

☐ stb ☐ pz ☐ sp ☐ envip

☒ pss ☐ pac ☐ pstb ☐ pnoise

☐ pxf ☐ psp ☐ qpss ☐ qpac

☐ qpnoise ☐ qpxf ☐ qpss ☐ hb

☐ hbac ☐ hbnoise ☐ hbss

Periodic Steady State Analysis

Engine ☒ Shooting ☐ Harmonic Balance

Fundamental Tones

#	Name	Expr	Value	Signal	SrcId
2		$1/(1u-0)$	1M	Large	V2
1		$1/(1u-0)$	1M	Large	V1

Large ☒

Clear/Add Delete Update From Hierarchy

☒ Beat Frequency ☐ Beat Period 1M Auto Calculate ☒

Output harmonics

Number of harmonics ☒ 6

Accuracy Defaults (errpreset)

☐ conservative ☒ moderate ☐ liberal

Transient-Aided Options

Run transient? ☒ Yes ☐ No ☐ Decide automatically

Detect Steady State ☒ Stop Time (tstab) 5u

Save Initial Transient Results (saveinit) ☐ no ☒ yes

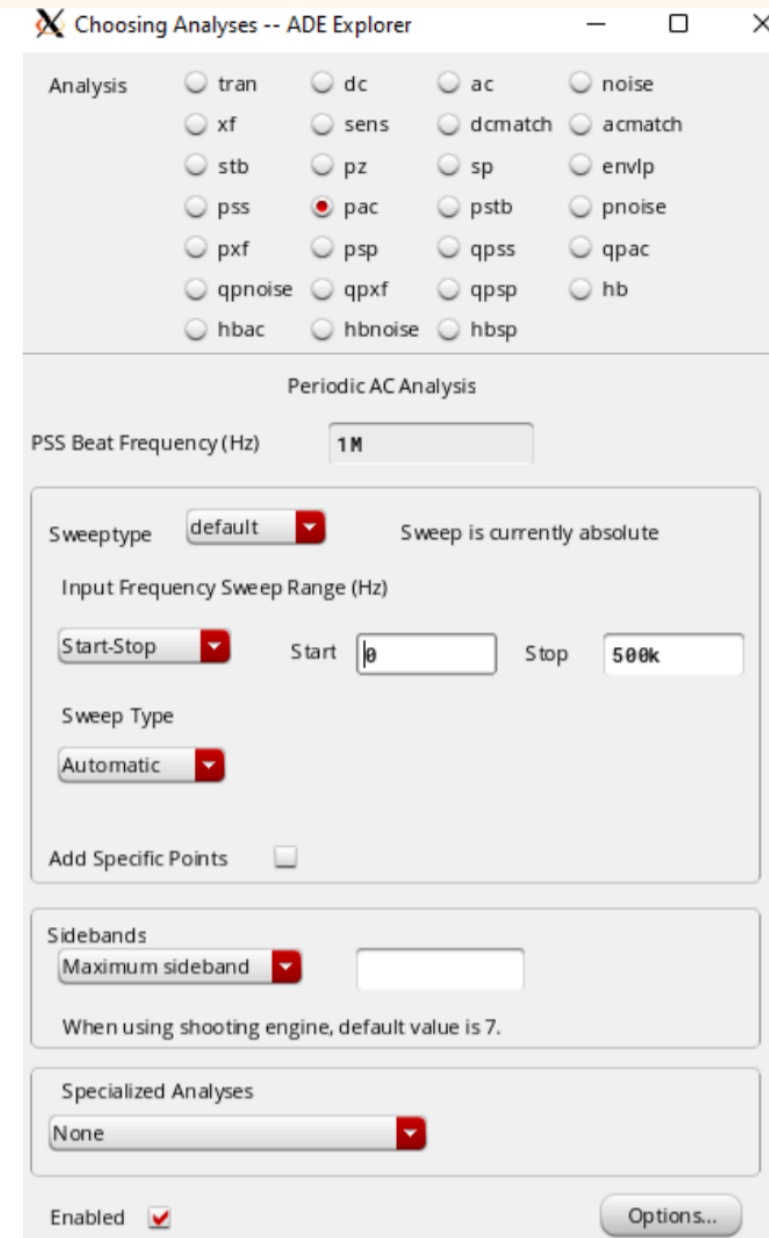
- DC is needed to check bias
- But some circuit is time varying, the bias changes, this is hard to simulate, need to transient all the way
 - but if the time varying is periodic, then is simpler, this is periodic steady state, just run one cycle
- Good for switch cap
- periodic steady state only need one cycle, but might need time to reach the steady state, so click the run transient to make it reach it properly
- Tutorial never go through how to see the bias though

- the DC op annotation also dh one
- seems to be a transient plot thing only, since its supposed not just be two states, bu a continues change in state that repeats

Setup PAC Simulation

Setup PAC Simulation

- Select [Analyses->Choose...](#)
- Choose pac analysis
- Define “Start-Stop” from 0Hz to 500kHz. Stop frequency should be half of sampling clock.
- As we did not define the “Maximum sideband”, the simulation will show up to 7 sidebands as follows:
 - 0: 0-500kHz
 - -1: 500k-1MHz
 - +1: 1MHz-1.5MHz
 - ...
 - -6: 5.5MHz-6MHz
 - +6: 6-6.5MHz
 - -7: 6.5-7MHz
 - +7: 7-7.5MHz

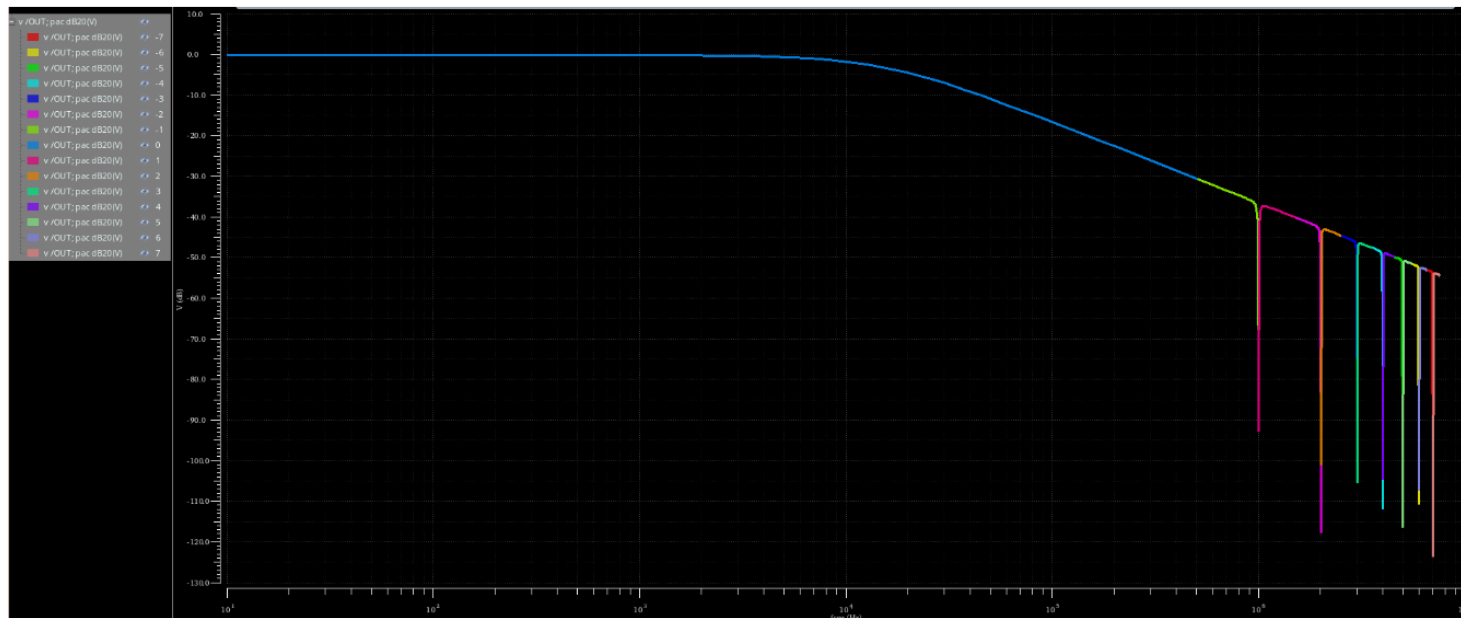


- what we wanted was the filter response
- so need PAC for AC response

- will run only up to Nyquist BW

Plotting PAC Simulation

- Select **Results->Direct Plot->Main Form...**
- Choose pac
- Select “Voltage”, “spectrum” and “dB20”. Then select the net that you want to probe from the schematic window
- Note that the different colour covers different harmonic, i.e. frequency range.



- its a LPF
- check the cutoff should be similar to the theory

Plotting Mode **Append**

Analysis

☐ pss ☒ pac ☐ tstab

Function

☒ Voltage ☐ Voltage Gain
☐ Current ☐ IPN Curves

Select **Net**

Sweep

☒ spectrum ☐ sideband

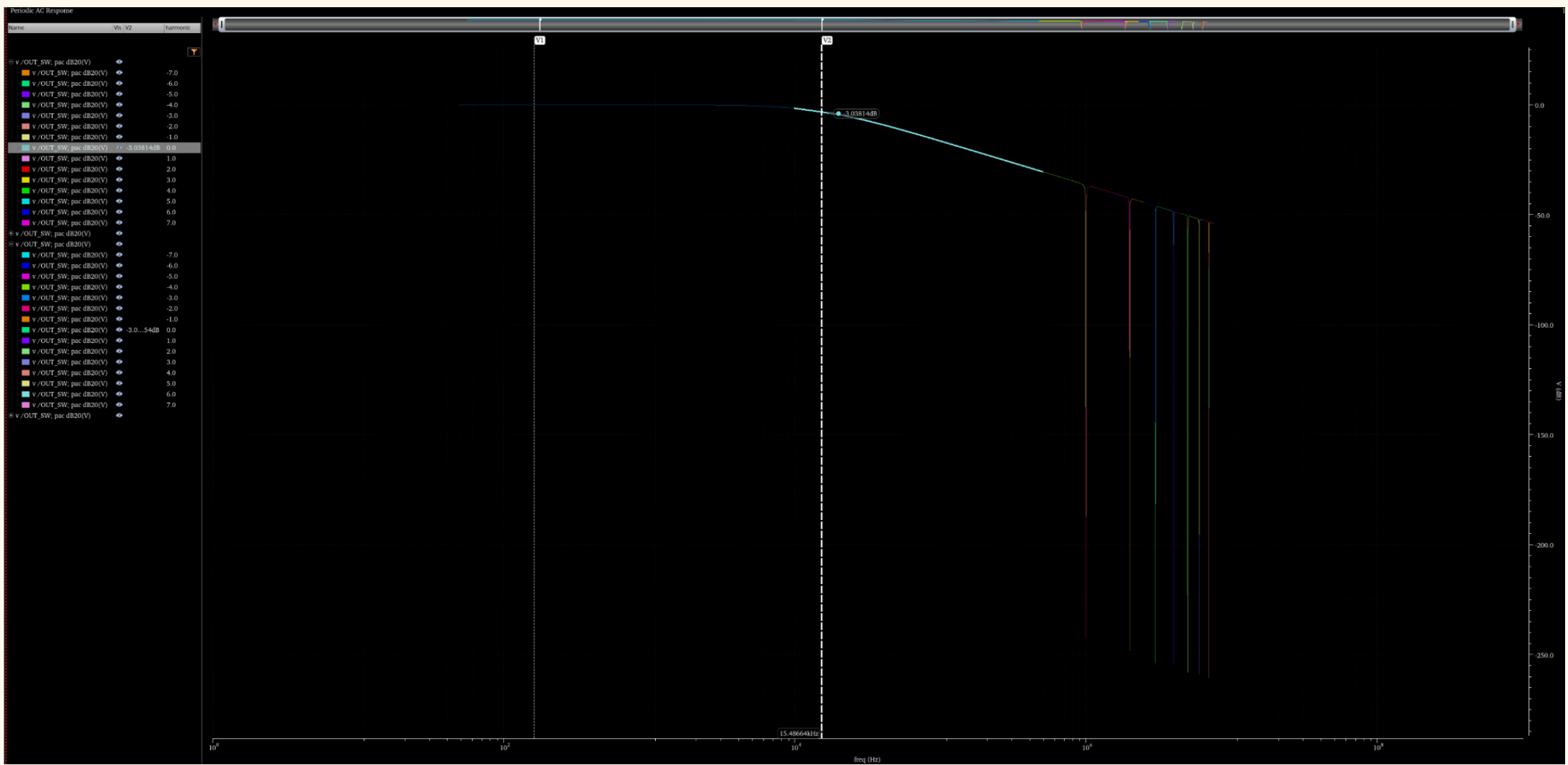
Modifier

☐ Magnitude ☐ Phase ☒ dB20
☐ Real ☐ Imaginary

Add To Outputs ☐ Replot

freqaxis = absout
 > Select Net on schematic...

OK **Cancel** **Help**



- the MOS and ideal switch overlaps each other
- 3dB $\approx$ 15.5 kHz
 - by theory it is

1	$\frac{1}{2\pi RC_2}$	$\times$
	=	15915.4943092
2	$R = \frac{T}{C_1}$	$\times$
	=	1000000
3	$T = 1 \cdot 10^{-6}$	$\times$
	=	0.000001
4	$C_1 = 1 \cdot 10^{-12}$	$\times$
	=	1×10^{-12}
5	$C_2 = 10 \cdot 10^{-12}$	$\times$
	=	1×10^{-11}

■

■ not bad

Setup PNOISE Simulation

- Select [Analyses->Choose...](#)
- Choose pnoise analysis
- Define “Start-Stop” from 0Hz to 500kHz. Stop frequency should be half of sampling clock.
- Define the output node to observe the noise voltage, which are “/OUT” and “gnd!” respectively

Choosing Analyses -- ADE Explorer

Analysis

☐ tran	☐ dc	☐ ac	☐ noise
☐ xf	☐ sens	☐ dcmatch	☐ acmatch
☐ stb	☐ pz	☐ sp	☐ envlp
☐ pss	☐ pac	☐ pstb	☒ pnoise
☐ pxf	☐ psp	☐ qpss	☐ qpac
☐ qpnoise	☐ qpxf	☐ qpssp	☐ hb
☐ hbac	☐ hbnoise	☐ hbbsp	

Periodic Noise Analysis

PSS Beat Frequency (Hz)

Multiple noise ☐

Sweep type Sweep is currently absolute

Output Frequency Sweep Range (Hz)

Start Stop

Sweep Type

Add Specific Points ☐

Sidebands

Method ☒ default ☐ full spectrum

When using shooting engine, default value is 7.

Noise Figure ☐

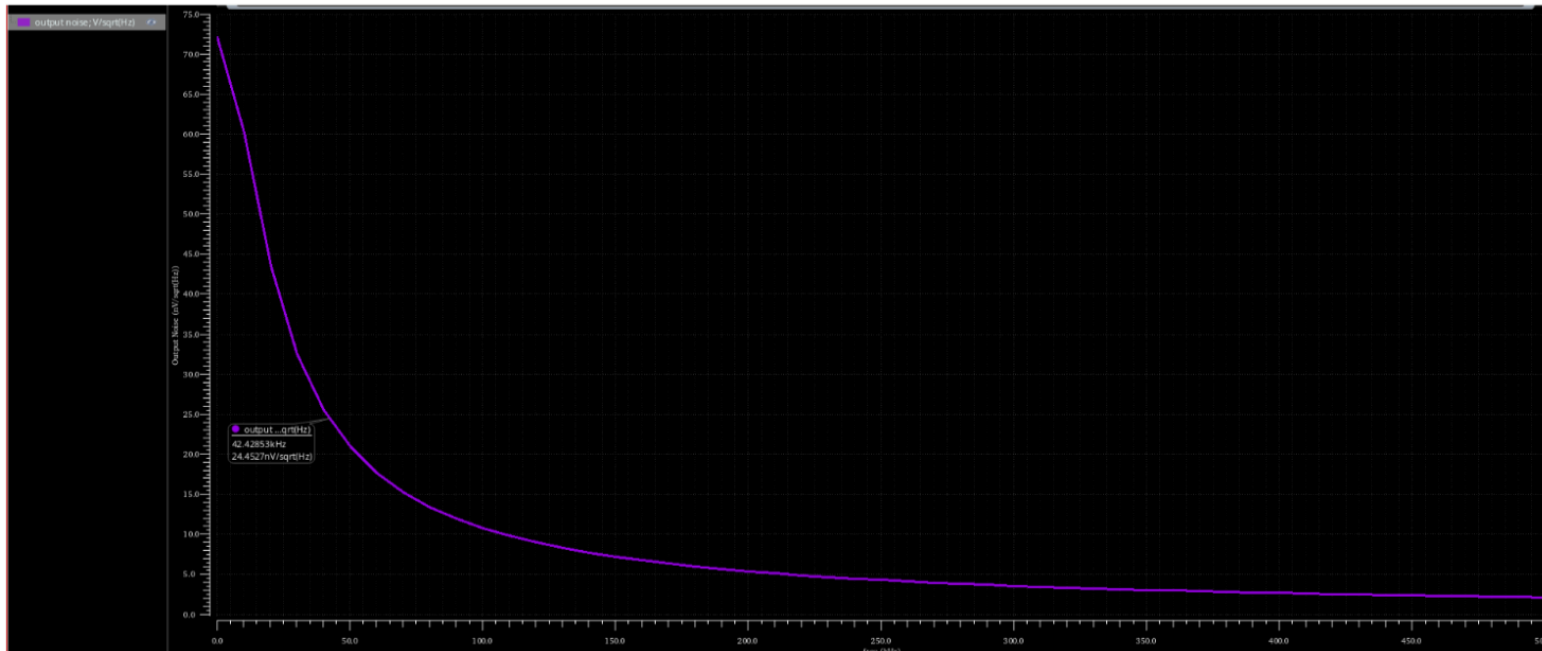
Output

Positive Output Node

Negative Output Node

- the current noise component of the mosfet will be changed based on the gate bias, so it will change periodically also, so need PNOISE

- Select **Results->Direct Plot->Main Form...**
- Choose pnoise
- Select “Output Noise”, “V/sqrt(Hz)” and plot
- The output noise spectrum from 0-500kHz is shown.
- As illustrated, at 42.42kHz, the noise voltage spectral is $24.45\text{nV}/\sqrt{\text{Hz}}$



Plotting Mode **Replace**

Analysis

☐ pss ☐ pac ☒ pnoise ☐ tstab

Function

☒ Output Noise ☐ Input Noise
☐ Noise Figure ☐ Noise Factor
☐ NFdsb ☐ Fdsb
☐ NFtee ☐ Ftee
☐ Transfer Function

Units

☒ V/sqrt(Hz) ☐ V^2/Hz ☐ dBc/Hz ☐ dBV/Hz

Loadpull Contours ☐

Add To Outputs ☐ **Plot**

> Press plot button on this form...

OK **Cancel** **Help**

Printing Noise Summary

- Select **Results->Print->Noise Summary...**
- Choose “integrated noise”
- Click “Include All Types”
- Under “Truncate & Sort”, enter “10”. This will summarize the top 10 noise contributors
- As illustrated, the NMOS drain noise current contribute the most to the total integrated output noise

Results Display Window

Window Expressions Info Help

cadence

Device	Param	Noise Contribution	% Of Total
/NM1	id	5.83314e-11	49.96
/NM0	id	5.82876e-11	49.92
/NM1	igs	4.41159e-14	0.04
/NM1	igd	4.37932e-14	0.04
/NM0	rd	1.35847e-14	0.01
/NM0	rs	1.35723e-14	0.01
/NM1	rs	1.26757e-14	0.01
/NM1	rd	1.26646e-14	0.01
/NM0	fn	8.2708e-16	0.00
/NM1	fn	5.9895e-16	0.00

Integrated Noise Summary (in V^2) Sorted By Noise Contributors
Total Summarized Noise = 1.16761e-10
No input referred noise available
The above noise summary info is for pnoise_usb data

Noise Summary

Print the output noise of `pnoise\_usb-(OUT gnd!)` analysis

Type ☐ spot noise ☒ integrated noise noise unit **V^2**

From (Hz) **1K** To (Hz) **500K**

weighting ☒ flat ☐ from weight file

FILTER

hierarchy level

bsim4
bsource_08102a

include instances

exclude instances

TRUNCATE & SORT

truncate **by number** top **10**

sort by ☒ noise contributors ☐ composite noise ☐ device name

OK Cancel Apply Help

- can see the noise is mainly from the drain current of the switching transistor

Noise

- You can perform “noise” or “pnoise” simulation similarly
- If ideal “res” components are chosen from “analoglib”, you need to indicate “Generate noise?” to be “yes” before the noise analysis can be performed
- Noise or pnoise analysis are useful for us to identify critical components that impact the output noise and whether the resulting output noise are acceptable

- this can be used to isolate the noise source
 - turn of generate noise

Add Instance

Library:

Cell:

View:

Names:

☒ Add Wire Stubs at:
☐ all terminals ☒ registered terminals only

Array: Rows Columns

Model name:

Resistance:

Length:

Width:

Multiplier:

Scale factor:

Temp rise from ambient:

Temperature coefficient 1:

Temperature coefficient 2:

Resistance Form:

Generate noise? ☒

Periodic vs Non-Periodic Simulation

- PSS, PAC, PNOISE are equivalent DC, AC and NOISE simulation performed on circuits that have periodic steady state, for example, switched capacitor, oscillator, mixer and etc.
- For normal analog circuit, it is usually biased at certain DC operating point, and the AC and NOISE behavior can be analyzed surrounding the DC operating point.
- For RF or switched capacitor circuit, they will have different steady state point depending on the time instant, usually related to the clock cycle the circuit experienced. Hence, PSS, PAC and PNOISE need to be performed, where the solution is found over one cycle of the clock period